

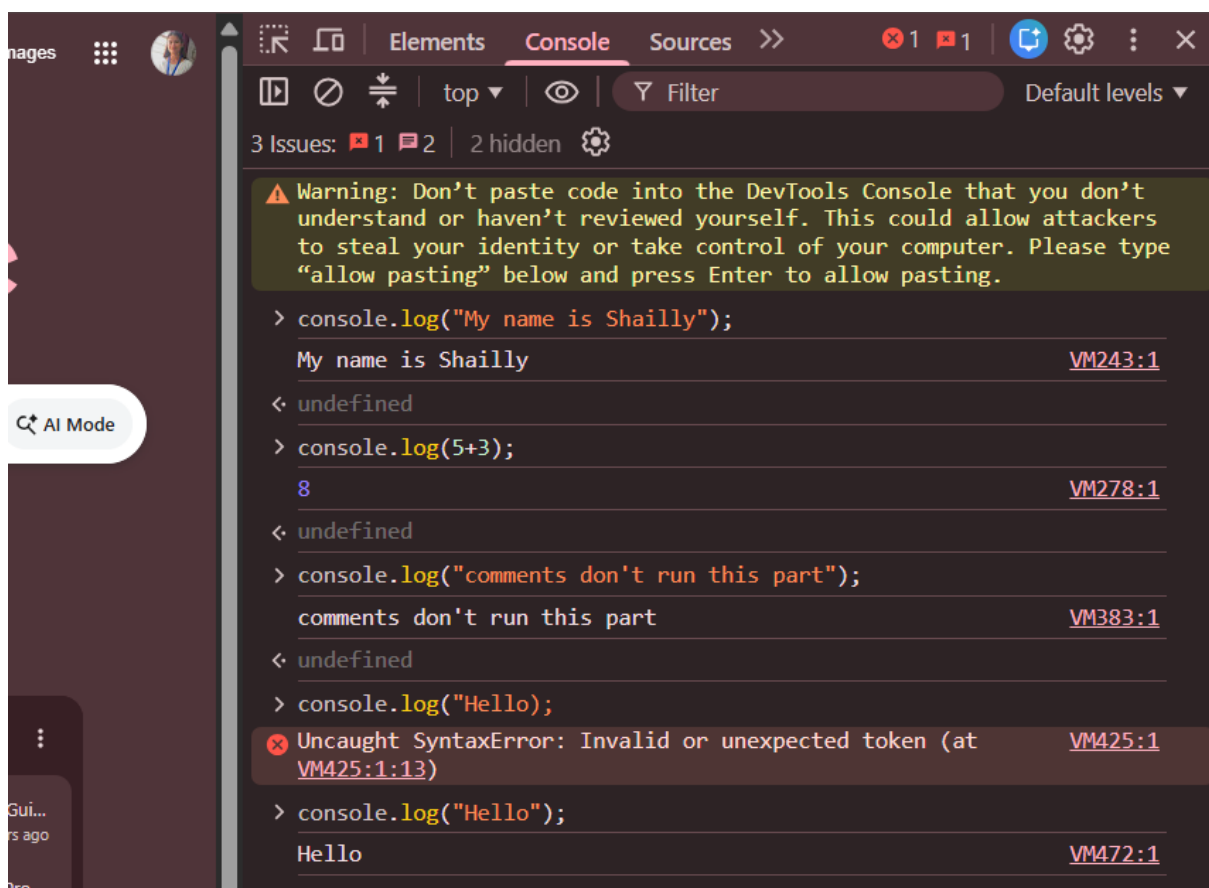
Name = Shailly
course = BCA 5Th sem

Task 1.1:

Predicted: Syntax error because the closing quotation mark is missing.

Actual: SyntaxError was displayed.

Correction: Added the missing closing quotation mark.



Task 1.

```
< undefined
> let test = "Hello";
  console.log(Test);
✖ ▶ Uncaught ReferenceError: Test is not defined      VM600:2
   at <anonymous>:2:13
> |
```

Answer:- JavaScript is case-sensitive, so test and Test are treated as different identifiers.

Task 2.1

```
< undefined
> const collegeName = "Dev Sanskriti Vishwavidyalaya";
< undefined
> collegeName = "ABC college";
✖ ▶ Uncaught TypeError: Assignment to constant variable.      VM1666:1
   at <anonymous>:1:13
>
```

Task 2.3

Code	Answer Reason
let 1stYear;	Invalid A variable name cannot start with a number.
let student_name;	Valid Underscore _ is allowed in variable names.
let let;	Invalid let is a reserved JavaScript keyword.
let \$marks;	Valid Dollar sign \$ is allowed in variable names.
let roll no;	Invalid Spaces are not allowed in variable names.

Task 2.4

```
> if (true){
    var a = "I am var";
    let b = "I am let";
}
console.log(a);
console.log(b);

I am var VM1930:5
✖ ▶ Uncaught ReferenceError: b is not defined VM1930:6
    at <anonymous>:6:13
```

Explanation:

var can be accessed outside the block, but let is block-scoped, so b cannot be accessed outside the if block

Task 3.2

```
> let myString = "Shailly";
let myNumber = 22;
let myBoolean = true;
let myUndefined;
let myNull = null;
let myBigInt = 12345678901234567890;
let mySymbol = Symbol("id");

console.log(myString, typeof myString);
console.log(myNumber, typeof myNumber);
console.log(myBoolean, typeof myBoolean);
console.log(myUndefined, typeof myUndefined);
console.log(myNull, typeof myNull);
console.log(myBigInt, typeof myBigInt);
console.log(mySymbol, typeof mySymbol);

Shailly string VM3986:9
22 'undefined' VM3986:10
true 'boolean' VM3986:11
undefined 'undefined' VM3986:12
null 'object' VM3986:13
12345678901234567000 'number' VM3986:14
Symbol(id) 'symbol' VM3986:15
< undefined
> const student = {
    name: "Shailly",
    rollNo: 2424334,
    course: "BCA",
    ispassing: true
};
console.log(student.name);
console.log(student.isPassing);
const subjects = ["JavaScript", "DBMS", "Networking", "web Development"];

console.log(subjects[0]);
console.log(subjects[3]);

Shailly VM4911:7
```

```

> const student = {
  name: "Shailly",
  rollNo: 2424334,
  course: "BCA",
  ispassing: true
};
console.log(student.name);
console.log(student.isPassing);
const subjects = ["JavaScript", "DBMS", "Networking", "web Development"];

console.log(subjects[0]);
console.log(subjects[3]);

```

Shailly

undefined

JavaScript

web Development

< undefined

Task 3.3

```

> console.log(typeof null);

```

object

< undefined

I predicted object, and the actual output was also object. This is a historical behavior of JavaScript.

Task 4.1

```

> let x = 20;
let y = 6;
console.log("sum:",x+y);
console.log("Difference:",x-y);
console.log("product:",x*y);
console.log("Quatient:",x/y);
console.log("Remainder:",x%y);

```

sum: 26

Difference: 14

product: 120

Quatient: 3.3333333333333335

Remainder: 2

< undefined

Task 4.2

== can convert the data type before comparison, while **===** checks both value and data type without type conversion.

Task 4.3

Task 4.4

Part 5 —

```
> let age = 17;
let hasID = true;

console.log(age >= 18 && hasID);
console.log(age >= 18 || hasID);
console.log(!hasID);

false
true
false
< undefined
> |
```

```
> let marks = 75;
let result = marks >= 40 ? "Pass" : "Fail";
console.log(result);
marks = 30 ;
result = marks >= 40 ? "Pass" : "Fail";
console.log(result);

Pass
Fail
< undefined
```

The screenshot displays a web browser window on the left and a code editor on the right. The browser window shows a web page titled "JavaScript Grade Calculator" with a console output area. The console output shows the following results:

- Total: 255
- Average: 85
- Grade: B
- Average data type: number
- Scholarship Eligible: true
- Total: 255, Average: 85.00, Grade: B
- Live reload enabled.

The code editor on the right shows the JavaScript code for the grade calculator. The code is as follows:

```
1 let subject1 = 85;
2 let subject2 = 78;
3 let subject3 = 92;
4
5 let attendance = 80;
6
7 let total = subject1 + subject2 + subject3;
8 let average = total / 3;
9
10 let grade = average >= 90 ? "A" :
11     average >= 75 ? "B" :
12     average >= 40 ? "C" : "F";
13
14 let isEligibleForScholarship =
15     average >= 85 && attendance >= 75;
16
17 console.log("Total:", total);
18 console.log("Average:", average);
19 console.log("Grade:", grade);
20 console.log("Average data type:", typeof average);
21 console.log("Scholarship Eligible:", isEligibleForScholarship);
22
23 console.log(
24     "Total: " + total +
25     ", Average: " + average.toFixed(2) +
26     ", Grade: " + grade
27 );
```